

Deep Learning Specialization

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ANN $\rightarrow$ Supervised Learning problems.

Rule:

Activation
function for
output layer in ANN

function that introduce non-linearity
behavior in the output.

Regression

number of units: 1

activation : linear
(pass-through)
 $f(x) = x$

Classification

Sigmoid
softmax

Binary
Classification

number of units: 1

activation : sigmoid

-or-

number of units: 2

activation : softmax

Multi-class
Classification

number of units : no. of unique
labels in
label column.

activation : softmax

eg: iris dataset
Setosa, Ve, Vi°

units : 3

act : softmax

Guidelines

Activation functions for Hidden layer in ANN

sigmoid

tanh
(hyperbolic
tangent)

Avoid if you are dealing
with deep NN.

VANISHING GRADIENT DESCENT
 Δw & Δb

$\Delta w \sim 0$
 $\Delta b \sim 0$

No learning
No improvement

relu
(Rectifier
linear
unit)

first preference
 $\sim 90\%$ problems
solved

if $x > 0$:
 $f(x) = x$
else:
 $f(x) = 0$

leakyrelu
second
preference

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